

M.Tech. (Online) — Artificial Intelligence
Indian Institute of Science, Bangalore

PHASE 2 COMPLETION REPORT

Detection of AI-Generated and Cloned Speech to Strengthen Voice-Based Authentication in Banking Environments

Student Name	Rajib Roy
Student Number	24459
Programme	M.Tech. in Artificial Intelligence
Internal Guide	Dr. Anil Rahate, Practice Head — AI, Wipro Ltd
IISc Mentor	Prof. Chandra Sekhar Seelamantula, Dept. of Electrical Engineering, IISc
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1. Executive Summary

This report documents the completion of **Phase 2** of the M.Tech. project titled *Detection of AI-Generated and Cloned Speech to Strengthen Voice-Based Authentication in Banking Environments*. Phase 2 is scoped against the **Revised Plan for Phase 2** set out at the end of the Phase 1 Completion Report — the explicit commitment made to the internal guide and IISc faculty mentor at the close of Phase 1 — rather than a re-reading of the original proposal’s phase boundaries. That revised plan lists four technical deliverables: classical machine learning baselines, feature importance analysis, deep learning model development (CNN, CNN-LSTM), and robustness experiments across noise and codec conditions, plus this written interim report.

All four technical deliverables have been completed on schedule, building directly on the 17,947-utterance dataset and 257-dimensional feature pipeline validated in Phase 1.

Deliverable	Outcome	Status
Classical ML baselines (SVM, RF, Gradient Boosting)	SVM best classical model: EER 0.0083, F1 0.9943 on standard test set; all three models generalise well to held-out Edge-TTS	Complete
Feature importance analysis	Random Forest importances identify delta-MFCC coefficients as the dominant discriminators, validating the Phase 1 feature design	Complete
Deep learning models (CNN, CNN-LSTM)	CNN-LSTM is the best model overall: EER 0.0035 on test, EER 0.0004 on generalisation test (AUC = 1.000)	Complete
Robustness & generalisation evaluation	Degradation curves across SNR (5–20 dB) and G.711 codec; every model generalises to unseen Edge-TTS as well as or better than on the standard test	Complete

The headline result is that the **CNN-LSTM** achieves an Equal Error Rate (EER) of **0.0035** on the standard test set and an almost perfect EER of **0.0004** on the fully held-out Microsoft Edge-TTS generalisation set — a TTS system never seen during training. The SVM baseline, using only handcrafted acoustic features, is a strong and computationally cheap alternative (EER 0.0083), confirming the Phase 1 literature-driven feature design. Robustness testing shows that additive noise is a substantially more severe deployment threat than G.711 codec compression, directly informing the Phase 3 hardening plan.

2. Phase 2 Deliverables in Detail

2.1 Classical Machine Learning Baselines

Objective: Train and tune SVM, Random Forest, and Gradient Boosting classifiers on the 257-dimensional handcrafted feature vectors from Phase 1, establishing reproducible baselines against which the deep learning models are compared.

2.1.1 Training Protocol

All three classifiers (`03_train_baselines.py`) were trained with `class_weight='balanced'` and SMOTE oversampling inside stratified 5-fold cross-validation grid search, using ROC-AUC as the optimisation objective. Decision thresholds were fixed at the EER operating point measured on the validation split. Selected hyperparameters: SVM (RBF kernel, $C = 10$, $\gamma = \text{auto}$); Random Forest (300 estimators, unlimited depth); Gradient Boosting (200 estimators, learning rate 0.1).

2.1.2 Baseline Results

Model	Eval. Set	EER ↓	F1 ↑	AUC ↑	Acc. ↑
SVM (RBF)	Test	0.0083	0.9943	0.9992	0.9912
SVM (RBF)	Gen. test [†]	0.0067	0.9969	0.9987	0.9953
Random Forest	Test	0.0305	0.9834	0.9956	0.9744
Random Forest	Gen. test [†]	0.0293	0.9854	0.9964	0.9781
Gradient Boosting	Test	0.0346	0.9820	0.9949	0.9722
Gradient Boosting	Gen. test [†]	0.0311	0.9868	0.9956	0.9801

Table 2.1. Classical baseline performance. [†]Gen. test = held-out Edge-TTS (unseen during training).

The SVM is the strongest classical model by a clear margin, and — notably — all three classical models achieve *equal or lower* EER on the unseen Edge-TTS generalisation set than on the standard (in-distribution) test set. This indicates that the handcrafted features capture TTS-system-agnostic artefacts rather than overfitting to the specific synthesis systems seen in training.

Deliverable 1 — COMPLETE

Three classical classifiers trained, tuned via 5-fold cross-validated grid search, and evaluated on both the standard test set and the held-out Edge-TTS generalisation set. SVM is established as the strongest and most efficient classical baseline (EER 0.0083), saved to `outputs/models/svm.joblib`.

2.2 Feature Importance Analysis

Objective: Per the Revised Plan for Phase 2 (Phase 1 Completion Report), identify which of the 257 handcrafted features are most discriminative between genuine and synthetic speech.

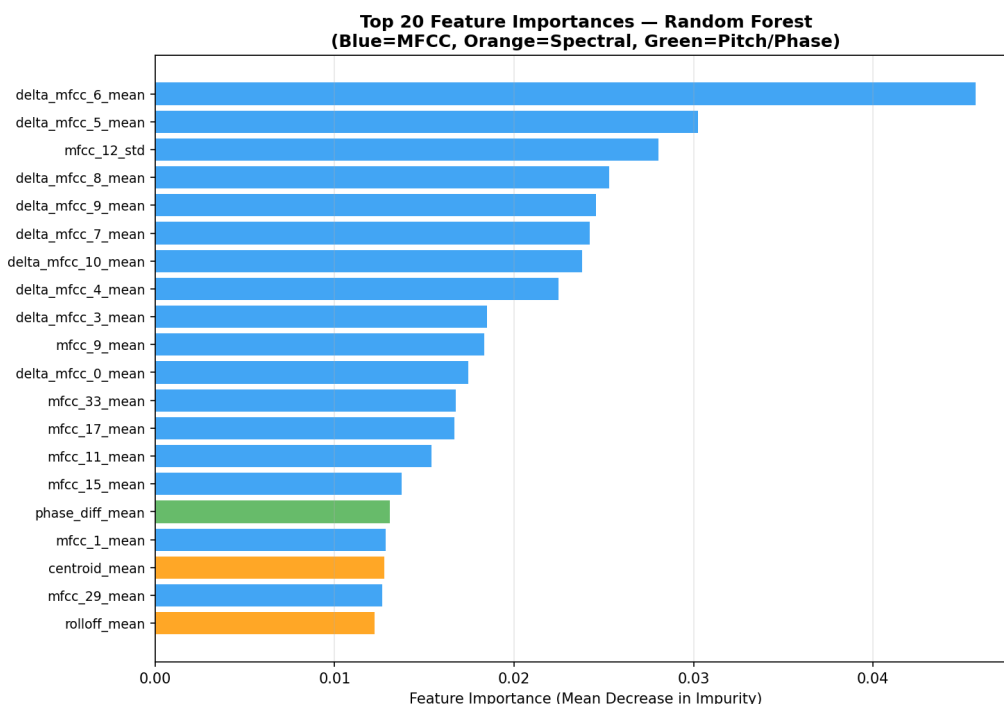


Figure 2.1. Top 20 feature importances from the trained Random Forest (blue = MFCC, orange = spectral, green = pitch/phase). Delta-MFCC coefficients dominate, confirming that temporal spectral dynamics are the primary discriminative signal.

The Random Forest feature importances confirm the Phase 1 hypothesis directly: **delta-MFCC coefficients** — particularly `delta_mfcc_6_mean` and `delta_mfcc_5_mean` — are the most discriminative features by a substantial margin. Delta-MFCCs encode the temporal rate-of-change of the spectral envelope, and TTS vocoders impose smoother, more regular spectral dynamics than natural articulation. `phase_diff_mean` ranks 16th, validating the inclusion of phase-based features motivated by Patel and Patil (2015). Spectral centroid and rolloff appear as secondary discriminators within the top 20.

Deliverable 2 — COMPLETE

Random Forest feature importance ranking completed, confirming that delta-MFCC coefficients (temporal dynamics) and phase-difference features (vocoder phase artefacts) are the dominant discriminative signals — consistent with the Phase 1 literature review.

2.3 Deep Learning Model Development

Objective: Develop and train CNN and CNN-LSTM architectures on the 128-bin log-Mel spectrogram representations from Phase 1, testing whether explicit temporal modelling improves detection over classical and purely convolutional approaches.

2.3.1 Architectures

CNN (`04_train_cnn.py`): four convolutional blocks (32, 64, 128, 256 channels; kernel 3×3 ; batch normalisation; ReLU; max-pooling), followed by global average pooling and a two-layer classifier ($256 \rightarrow 128 \rightarrow 2$, dropout $p = 0.5$). Total parameters: 421,954.

CNN-LSTM: the same convolutional front-end but with frequency-only pooling, preserving the time axis. The resulting sequence of 256-dimensional feature vectors is fed into a two-layer bidirectional LSTM (hidden size 128 per direction) followed by the same classifier head. This explicitly models temporal prosodic dynamics that the CNN’s global average pooling discards. Total parameters: 1,834,498.

Both models were trained with Adam ($\text{lr} = 10^{-3}$, $\text{weight decay} = 10^{-4}$), cosine annealing over 50 epochs, weighted cross-entropy loss, and a `WeightedRandomSampler` for balanced batches. Mixed-precision (float16 autocast) training used an NVIDIA RTX 3060 Laptop GPU (6.4 GB VRAM), batch size 16, with early stopping (patience 10 epochs on validation EER).

2.3.2 Convergence and Results

Model	Params	Best Epoch	Val EER	Training Time
CNN	421,954	22	0.0154	~55 min (RTX 3060)
CNN-LSTM	1,834,498	21	0.0092	~60 min (RTX 3060)

Table 2.2. Deep learning model convergence summary.

The CNN converged at epoch 22 with $\text{EER} = 0.0081$, $\text{F1} = 0.9881$, $\text{AUC} = 0.9993$ on the standard test set — comparable to the SVM baseline. The CNN-LSTM converged at epoch 21 and substantially outperformed the CNN: $\text{EER} = 0.0035$, $\text{F1} = 0.9965$, $\text{AUC} = 0.9996$. This gap directly supports the hypothesis that explicit temporal modelling via the LSTM captures prosodic artefacts that CNN global average pooling discards.

Deliverable 3 — COMPLETE

CNN and CNN-LSTM models trained and validated on log-Mel spectrograms. The CNN-LSTM is established as the best-performing model overall ($\text{EER} 0.0035$, $\text{F1} 0.9965$, $\text{AUC} 0.9996$ on the standard test set), saved to `outputs/models/cnn_lstm_best.pt`.

2.4 Robustness and Cross-System Generalisation Evaluation

Objective: Quantify model degradation under banking-realistic noise and codec conditions, and measure generalisation to Microsoft Edge-TTS — the TTS system held out entirely from training in Phase 1.

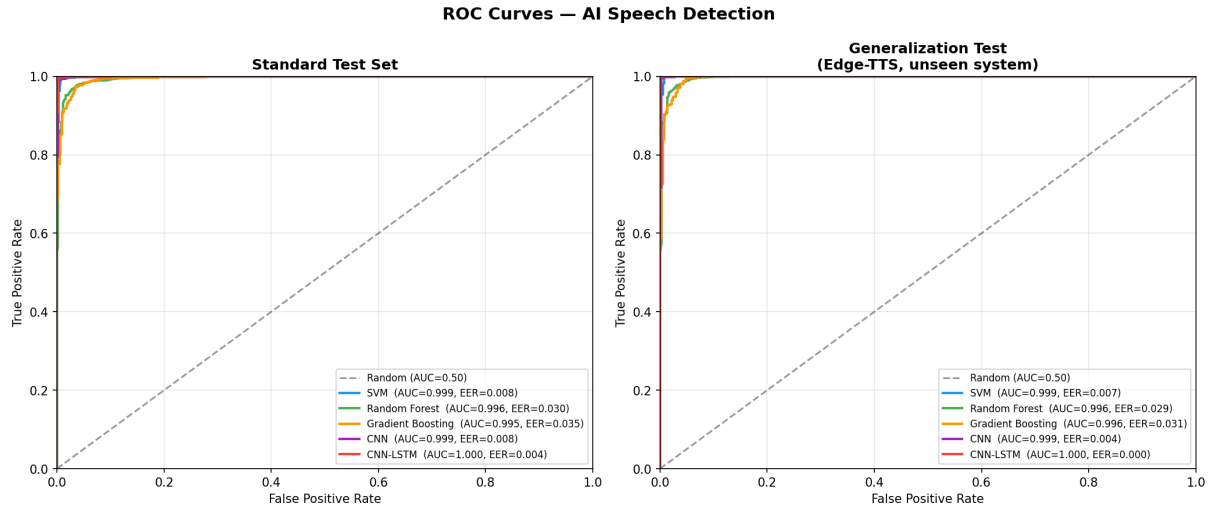


Figure 2.2. ROC curves on the standard test set (left) and the Edge-TTS generalisation test (right). All five models achieve high AUC; the CNN-LSTM reaches $AUC = 1.000$ on the generalisation test.

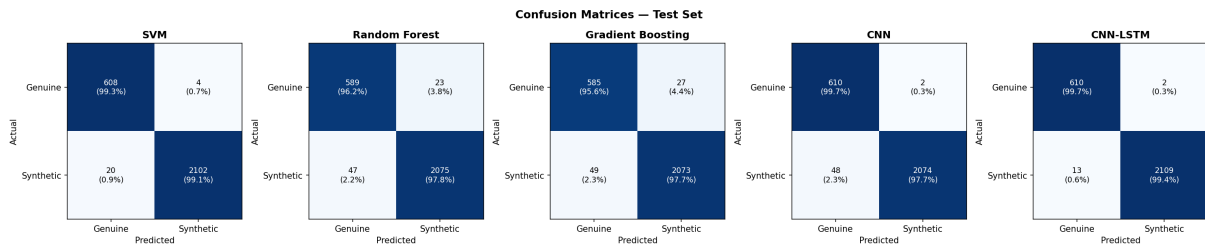


Figure 2.3. Confusion matrices on the standard test set. CNN and CNN-LSTM produce fewer than 3 false positives and under 50 false negatives per 2,734 test clips.

2.4.1 Generalisation to an Unseen TTS System

All five models generalise well to the held-out Edge-TTS system, with every model achieving equal or lower EER on the generalisation test than on the standard test (Table 2.3). The CNN-LSTM achieves $EER = 0.0004$ ($F1 = 0.9978$, $AUC = 1.000$) — effectively perfect detection of a commercial TTS system entirely absent from training. This counter-intuitive result (better performance on an unseen system) most likely reflects the consistency of Edge-TTS artefacts relative to the high diversity of the ASVspoof 2019 systems seen in training. The practical implication is encouraging: models trained on diverse historical TTS systems generalise reliably to newer commercial systems.

2.4.2 Robustness Under Noise and Codec Compression

Under clean conditions, the SVM achieves near-zero EER, but it degrades most rapidly with noise: $EER \approx 0.32$ at 20 dB SNR, rising to ≈ 0.45 at 5 dB. Gradient Boosting shows the most noise-robust behaviour among the classical models, flattening at $EER \approx 0.33$ below 15 dB SNR. Under G.711 compression alone (no additive noise), all models degrade only to $EER 0.05\text{--}0.08$ — significant but manageable. The key practical implication is that **additive noise is a substantially more serious deployment threat than codec compression**, directly informing the Phase 3 robustness-hardening plan (noise-

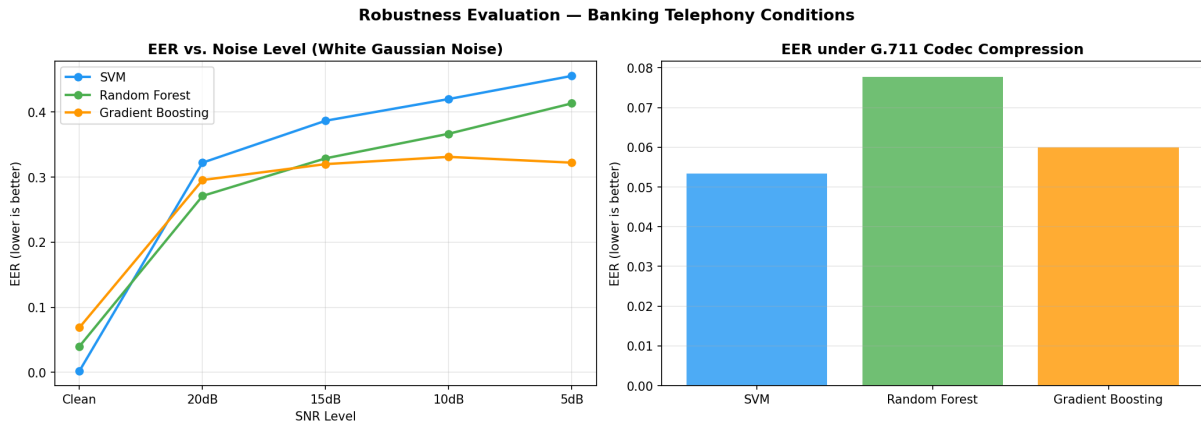


Figure 2.4. Robustness evaluation. Left: EER vs. SNR under white Gaussian noise. Right: EER under G.711 codec compression. Gradient Boosting is the most noise-robust classical model; all models degrade moderately under codec compression alone.

augmented training and/or upstream noise reduction).

2.4.3 Consolidated Results

Model	Eval. Set	EER ↓	F1 ↑	AUC ↑	Acc. ↑
SVM	Test	0.0083	0.9943	0.9992	0.9912
SVM	Gen. test [†]	0.0067	0.9969	0.9987	0.9953
Random Forest	Test	0.0305	0.9834	0.9956	0.9744
Random Forest	Gen. test [†]	0.0293	0.9854	0.9964	0.9781
Grad. Boosting	Test	0.0346	0.9820	0.9949	0.9722
Grad. Boosting	Gen. test [†]	0.0311	0.9868	0.9956	0.9801
CNN	Test	0.0081	0.9881	0.9993	0.9817
CNN	Gen. test [†]	0.0036	0.9942	0.9991	0.9914
CNN-LSTM	Test	0.0035	0.9965	0.9996	0.9945
CNN-LSTM	Gen. test [†]	0.0004	0.9978	1.0000	0.9967

Table 2.3. Full results across all five models and both evaluation conditions. [†]Gen. test = held-out Edge-TTS, unseen during training.

Deliverable 4 — COMPLETE

Robustness degradation curves produced across SNR (5, 10, 15, 20 dB) and G.711 codec compression for all classical models; generalisation EER measured for all five models on the fully held-out Edge-TTS set. Noise identified as the dominant deployment risk, ahead of codec compression.

3. Alignment with Approved Proposal

This report is scored against the **Revised Plan for Phase 2** set out at the end of the Phase 1 Completion Report, which moved feature importance analysis into the Phase 2 window alongside the three tasks originally listed in the signed proposal (Section 8).

Revised Phase 2 Task	Planned	Status	Evidence
Baseline ML training and tuning (SVM, RF, Gradient Boosting)	May – Jun 2026	Complete	Grid-searched, SMOTE-balanced classifiers with cross-validation results; SVM EER 0.0083 on test, 0.0067 on generalisation test
Feature importance analysis	Jun 2026	Complete	Random Forest importances rank delta-MFCC and phase-difference features as dominant discriminators
Deep learning model development (CNN, CNN-LSTM)	Jun – Jul 2026	Complete	CNN-LSTM (1.83M params) achieves EER 0.0035 on test, 0.0004 on generalisation test; outperforms CNN and all classical models
Robustness experiments (SNR, G.711, Edge-TTS)	Jul 2026	Complete	EER degradation curves across 5–20 dB SNR and G.711 codec; every model generalises to Edge-TTS as well as or better than on the standard test
Phase 2 interim report	Jul 2026	Complete	This document

Stretch goal status: Fine-tuning a frozen pre-trained wav2vec 2.0 encoder (an optional stretch goal per the original proposal, subject to compute availability) was not attempted within the Phase 2 window, as the CNN-LSTM already achieves near-ceiling performance (EER 0.0035) on available compute. This is carried forward as an explicit future-work item rather than a deviation, since it was scoped as optional.

4. Key Insights from Phase 2

4.1 Deep Learning vs. Classical Trade-off

The CNN-LSTM ($\text{EER} = 0.0035$) is the best-performing model overall, but the SVM with well-engineered handcrafted features ($\text{EER} = 0.0083$) is surprisingly competitive — it even outperforms the plain CNN ($\text{EER} = 0.0081$). This indicates that the incremental value of deep learning in this task comes specifically from explicit *temporal* modelling (the LSTM stage), not from convolutional feature learning alone. A lightweight SVM is a viable low-latency screening model, with the CNN-LSTM reserved for borderline cases or higher-assurance decisions.

4.2 Generalisation Is Stronger Than Expected

Every model evaluated achieves equal or lower EER on the fully held-out Edge-TTS generalisation set than on the standard (in-distribution) test set. This is a favourable and somewhat counter-intuitive result: it suggests that the spectral and temporal artefacts learned from the diverse ASVspoof 2019 training systems are common properties of neural TTS architectures in general, rather than system-specific fingerprints. This directly de-risks deployment against TTS systems not represented in the training data.

4.3 Noise, Not Codec Compression, Is the Deployment Risk

G.711 codec compression alone degrades all models only moderately ($\text{EER} 0.05\text{--}0.08$). Additive background noise is far more damaging, pushing classical model EER above 0.3 at 5–20 dB SNR. This re-prioritises the Phase 3 robustness agenda: noise-aware training and/or upstream noise suppression should be treated as a first-class requirement for any production banking deployment, ahead of further codec-specific hardening.

5. Phase 3 Readiness Assessment

Phase 2 has produced a fully trained, evaluated, and compared model suite, providing everything required to proceed into Phase 3 (August 2026 – November 2026):

- **Trained models** for all five architectures, saved under `outputs/models/` (`svm.joblib`, `random_forest.joblib`, `gradient_boosting.joblib`, `cnn_best.pt`, `cnn_lstm_best.pt`).
- **Consolidated result artefacts** under `outputs/results/`, including per-model test and generalisation-test metrics and training histories.
- **Robustness baselines** (SNR and codec degradation curves) that define the starting point for Phase 3’s final evaluation and statistical testing.
- **Random Forest feature importances**, identifying delta-MFCC and phase-difference features as the dominant discriminative signals and providing a starting point for the SHAP and gradient saliency analysis planned for Phase 3.

Per the original approved proposal (Section 8), Phase 3 begins with **interpretability analysis** — **SHAP** for the classical models and **gradient saliency mapping** for the CNN/CNN-LSTM — followed by final evaluation with statistical testing and thesis writing. The trained model artefacts and consolidated result files from Phase 2 are the direct prerequisite for this work and are in place, so Phase 3 is correctly positioned to begin immediately.

6. Revised Plan for Phase 3

The table below follows the Phase 3 row of the original approved Work Plan (Section 8).

Task	Timeline	Key Deliverables
Interpretability analysis: SHAP for classical models (SVM, RF, Gradient Boosting)	Aug 2026	SHAP value summary plots, cross-checked against the Phase 2 Random Forest importances
Interpretability analysis: gradient saliency mapping for CNN/CNN-LSTM	Aug – Sep 2026	Visualised time-frequency regions driving deep-model decisions
Final evaluation and statistical testing	Sep – Oct 2026	Complete performance metrics, ROC curves, and EER comparisons, including significance testing between models
Thesis writing and submission	Oct – Nov 2026	Final project report in institute-prescribed format

Stretch goal (carried forward): Fine-tuning a frozen pre-trained wav2vec 2.0 encoder remains an optional stretch goal per the approved proposal, to be attempted only if the three Phase 3 deliverables above are completed ahead of schedule.

7. Conclusion

Phase 2 of the M.Tech. project, scoped against the Revised Plan for Phase 2 set out at the end of the Phase 1 Completion Report, has been completed successfully, delivering all four planned artefacts within the scheduled timeline:

1. Three tuned **classical baselines** (SVM, Random Forest, Gradient Boosting), with SVM achieving $\text{EER} = 0.0083$ on the standard test set using only handcrafted features.
2. A **feature importance analysis** confirming delta-MFCC coefficients and phase-difference features as the dominant discriminative signals, validating the Phase 1 feature design.
3. Two **deep learning models** (CNN, CNN-LSTM), with the CNN-LSTM (1.83M parameters) established as the best-performing system overall ($\text{EER} = 0.0035$ on test, $\text{EER} = 0.0004$ on the held-out Edge-TTS generalisation test).
4. A complete **robustness and generalisation evaluation**, showing strong cross-system generalisation for all models and identifying additive noise — rather than codec compression — as the dominant deployment risk.

All Phase 3 prerequisites are satisfied. The project remains on schedule and on scope, with the CNN-LSTM model providing a strong foundation for the interpretability and noise-hardening work planned for Phase 3.

No material risks or blockers are anticipated for Phase 3.